Architectural Patterns for Integrating LLMs into User-Facing Applications

Lessons from a language-learning platform

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Large Language Models are increasingly being integrated as components into existing user-facing applications, alongside the more visible wave of LLM-native products such as chatbots and agents. The engineering concerns of the two cases differ: when an LLM is a component behind an existing feature rather than the product itself, designers must reconcile its per-token cost, multi-second latency, non-determinism, rapidly shifting provider landscape, and general-purpose capability with the expectations of users who already rely on the system. We present a catalogue of recurring architectural patterns that address these concerns, grounded in over a year of LLM integration work on Zeeguu, an open-source language-learning platform with several hundred monthly active users. The catalogue spans five concerns — cost optimization, latency and availability, lifecycle management, data management, and quality assurance — and is described using the standard pattern format. The patterns are presented as a starting point for community refinement and extension, not as a closed taxonomy.

Additional Key Words and Phrases: large language models, software architecture, design patterns, LLM integration, cost, latency, quality assurance

1 Introduction

While there is growing literature on building LLM-native products (chatbots, agents, RAG systems) and on using LLMs for code generation, there is surprisingly little guidance on the **software engineering challenges of integrating LLMs as components into existing interactive applications** where real users expect fast, reliable, and trustworthy responses.

We expect this integration to become the common case: over time, more and more existing user-facing applications will adopt LLMs not as standalone chatbots but as components working behind the scenes to improve the user experience.

LLMs have a unique combination of properties that create novel architectural forces:

- they are expensive (per-token pricing),
- slow (high latency),
- non-deterministic (same input can yield different outputs, although this can be to a certain degree controlled with the temperature setting),
- rapidly evolving (new models released every few months and old ones regularly deprecated),
- imprecise (they make mistakes), and
- general-purpose (they can attempt almost any task described as text).

These properties demand specific engineering strategies.

Over the past year, we have been integrating LLMs into Zeeguu, an open-source platform for personalized language learning that helps users learn foreign languages by reading authentic online content. Through this work, we have identified a set of recurring architectural patterns for LLM integration. We believe these patterns are general and applicable beyond our specific domain.

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2 Case Study: Zeeguu

Zeeguu is an open-source language learning platform built around the idea that learners benefit most from engaging with authentic content in their target language. Rather than relying on artificial textbook exercises, Zeeguu helps users find real articles (news, blog posts, and other web content) tailored to both 1) their *level* and 2) their *interests*.

The platform recommends articles in the learner's target language based on their proficiency and reading preferences, making it **easy to find material that is both engaging and appropriately challenging**. If a text is personally compelling but too difficult, Zeeguu simplifies it to the learner's level using LLMs.

When users encounter unfamiliar words or phrases while reading, they can get translations on the fly powered by (contextual) Google Translate, so the reading experience remains fluid and uninterrupted. One alternative is provided by a state-of-the-art LLM, offering a more contextually nuanced option.

Every translation a user requests is logged by the system, which over time builds a **detailed model of the learner's vocabulary knowledge**, tracking which words they know, which ones they struggle with, and how well they've retained previously encountered vocabulary.

Based on this evolving learner model, Zeeguu **generates interactive vocabulary exercises and audio lessons** that focus on the words that matter most for each individual learner, rather than following a generic curriculum. The exercises use the context in which the word was originally encountered, based on the assumption that if the original text was compelling to the learner, examples drawn from it will be too.

In essence, Zeeguu unifies reading, translation, learner modeling, and practice into a coherent pipeline, with the learner's own reading interests as the primary driver.

2.1 The Pieces

A handful of Zeeguu-specific concepts recur across the patterns. They are collected here once, so a pattern can point back to a single definition rather than re-explaining them.

2.1.1 CEFR Levels. The Common European Framework of Reference grades language proficiency on an ordered six-level scale, from A1 (beginner) to C2 (mastery). Zeeguu uses it in two directions: to estimate how hard an article is, and to simplify an article down to every level easier than the original.

2.1.2 Article Simplification. When an article is compelling but too hard, Zeeguu rewrites it with an LLM to easier CEFR levels, producing one simplified version per level below the original. It runs both on demand, when a reader opens an article that has not been simplified yet, and ahead of time, over the crawled feed.

2.1.3 Crawling. Zeeguu builds its article recommendations by crawling news sites and blogs multiple times per day; this steady feed of freshly crawled articles is where most ahead-of-time LLM work happens (CEFR assessment, simplification), off any reader's critical path. On demand, readers push their own content in: a browser extension sends any article to Zeeguu for study, and on mobile the system share sheet sends any web page to be made interactive.

2.1.4 Multi-Word Expressions. A multi-word expression (MWE) is a group of words whose meaning is not the sum of its parts, for example *break the ice*. Zeeguu detects them so a learner can translate the phrase as a unit rather than word by word. A cheap dependency-parse gate (Stanza) fires first, and an LLM confirms only the flagged sentences.

2.1.5 The Learner Model. Every translation a learner requests is logged, building a model of which words they know and which they struggle with. Each word-in-context is a *meaning*; meanings are classified (by frequency, CEFR level, and phrase type) and drive which exercises and lessons the learner sees.

2.1.6 Translation. While reading, a learner gets an instant contextual translation from Google Translate. If it reads poorly, they can escalate to an LLM through the "Ask AI" action for a more nuanced rendering.

2.1.7 Audio Lessons. Zeeguu generates personalized audio lessons: an LLM writes a short script built around the words a learner is studying, and text-to-speech synthesizes the audio. Both steps are slow and costly, so lessons are pre-computed for recently active learners.

2.1.8 Vocabulary Exercises. Exercises are built from the words a learner has looked up. They use example sentences drawn from the learner's own reading where possible, and LLM-generated, then LLM-validated, sentences otherwise.

2.2 Reach

Zeeguu currently serves over 300 monthly active users, with peaks exceeding 400 during the academic year[1]. It is publicly available to try: the web app is at zeeguu.org with the invite code `zeeguu-beta`, and the mobile apps can be downloaded from the iOS and Android app stores.

3 Cost Optimization

We use *cost* as shorthand for any expensive, metered resource: most visibly the per-token API bill. However, costs are also latency, compute, and energy. The unifying force is that LLM calls carry a large, often *fixed* overhead (the instructional prompt, the network round-trip), so invoking them naïvely wastes that setup on a tiny payload.

These patterns are really about *economy*, each attacking the bill from a different angle. *Prompt Amortization* **amortizes a fixed overhead** across many calls, so the expensive preamble is paid once rather than per item. *Escalate to the LLM* **spends the LLM only in proportion to the value it adds**, reaching for it only where a cheaper tool falls short. *Slow-Path Inference* **pays premium, low-latency rates only when a user is waiting**, routing deferrable work to a cheaper path. *Per-User Consumption Budget* **bounds what any single user can consume**, so user-driven demand cannot run the bill up without limit. The same frugality transfers to non-monetary resources, which is why Prompt Amortization cuts latency as well as dollars.

3.1 Prompt Amortization

3.1.1 Context. Many LLM calls share the same shape: a large, fixed instructional prompt (rules, taxonomies, output format, examples) wrapped around a tiny variable input (see figure). The work is offline and batchable: nobody is blocked on any single result.

3.1.2 Example. Several Zeeguu jobs have exactly this shape. Rather than pay the preamble once per item, they pack related items into a single call, in one of two directions: **fan-in** (many inputs, one call) or **fan-out** (one input, many outputs):

[1]Monthly active users are defined as users with any learning activity (exercises, reading, browsing, audio lessons, or translations) in a given month. Live statistics are available at: https://api.zeeguu.org/stats/monthly_active_users

```
COMBINED_VALIDATION_PROMPT = """You are validating and classifying a language learning translation.

A learner highlighted "{word}" in this {source_lang} sentence:
"{context}"

The translation is in {target_lang}: "{translation}"

CRITICAL: The translation "{translation}" is in {target_lang}, NOT English!
If {target_lang} is Dutch, German, etc., interpret the translation IN THAT LANGUAGE.
Example: "offer" in Dutch means "victim/sacrifice" - this is VALID for "victima" (Spanish).

STEP 1 - VALIDATION:
1. Is "{word}" a meaningful learning unit? (Not an arbitrary fragment like "bruger den" = "are using it")
2. Is "{translation}" a correct translation for the WORD itself (not the surrounding phrase)?
3. If the word is part of an idiom, decide: should the learner study the single word or the full idiom?

IMPORTANT: Only mark as FIX if the translation is WRONG. Do NOT "improve" correct translations.
- Keep translations SHORT and SIMPLE - learners must type them in exercises
- NO parenthetical explanations like "by (doing something)" - just "by"
- NO alternatives with "/" like "as/in the capacity of" - pick ONE: "as"
- NO articles unless essential: "eyes" not "the eyes"

STEP 2 - CLASSIFICATION (for the valid/corrected word):
Frequency:
- unique: only meaning of the word
- common: primary or frequently used meaning
- uncommon: infrequently used meaning
- rare: specialized, archaic, or context-specific

CEFR Level (what level learner should know this word):
- A1: basic everyday words (hello, cat, eat)
- A2: common everyday vocabulary (weather, shopping)
- B1: intermediate topics (opinions, work, travel)
- B2: abstract concepts, nuanced vocabulary
- C1: advanced, sophisticated vocabulary
- C2: rare, literary, highly specialized

Phrase type:
- single_word: individual word
- collocation: natural word combination ("strong coffee", "take place")
- idiom: non-literal meaning ("break the ice", "piece of cake")
- expression: common phrase/greeting ("how are you")
- arbitrary_multi_word: random fragment, NOT worth studying ("docter for", "the cat on", "bruger den")

Reply in this EXACT format (one line):
VALID|frequency|cefr|phrase_type|explanation|literal_meaning
or
FIX|corrected_word|corrected_translation|frequency|cefr|phrase_type|reason|explanation|literal_meaning

Fields:
- cefr: A1, A2, B1, B2, C1, or C2
- explanation: OPTIONAL usage notes, register (formal/informal). Leave empty if not needed.
- literal_meaning: ONLY for idioms - SHORT word-by-word translation (e.g., "kick into touch").
  Leave EMPTY for single words, collocations, and non-idioms. NOT for explanations!

Examples:
- Simple word: VALID|common|A1|single_word||
- Word with usage note: VALID|common|B1|single_word|formal register|
- Idiom: VALID|common|B2|idiom|kick into touch
- Wrong translation: FIX|ojne|ne the eyes|common|A2|single_word|literal meaning is 'the eyes'||
- Idiom fix: FIX|se virkelighedens i ojnene|face reality|common|B2|idiom|idiom meaning 'face reality'||see reality in the eyes
- Arbitrary fragment: FIX|bruger|ose|common|A2|single_word|'bruger den' is arbitrary fragment||
"""
```

Fig. 1. The `COMBINED_VALIDATION_PROMPT` template: ~250 lines of validation rules, frequency/CEFR/phrase-type taxonomies, output format, and examples, wrapped around just three variables (`{word}`, `{translation}`, `{context}`). Sent one pair at a time, the entire preamble is re-paid on every call. This fixed overhead is the cost the pattern amortizes.

- **Meaning classification** (*fan-in*) sends ~15 word-meanings per call, sharing one frequency/CEFR-type taxonomy prompt across the whole batch.[2]
- **Example-sentence validation** (*fan-in*) checks ~20 generated examples per call.[3]
- **Article simplification** (*fan-out*) produces every CEFR level simpler than the original in one call, one section per level, turning four or five requests into one, about 75% fewer calls for a typical article.[4]

3.1.3 Problem. Sent one item at a time, that fixed preamble is re-paid on every call and dominates token cost and latency. How can it be paid once instead of once per item, without blowing past the model's quality or context limits?

3.1.4 Forces.

- **Preamble overhead.** The preamble is large by necessity: detailed rules, taxonomies, and examples are what make the output accurate and consistent, so quality pulls it up, and it cannot be trimmed without losing quality. Sent one item at a time, that fixed preamble is re-paid on every call, in tokens, cost, and latency. The only lever left is to amortize it. *(pushes toward bigger batches)*
- **Quality ceiling.** Accuracy and consistency degrade as more items share one call; past ~15–20 small items some models start dropping or muddling entries. *(pushes toward smaller batches)*
- **Context ceiling.** Input *and* output must fit the window; for fan-out the output side binds first, since each result is full-length.
- **Interactive latency.** Fan-in requires waiting to accumulate enough items to fill a batch: fine offline, but unacceptable when a user is blocked on a single result.

[2]The batched meaning-classifier prompt, `create_batch_meaning_frequency_and_type_prompt` in the `zeeguu/api` repository.
[3]The batch example-sentence validator, `validate_examples_batch` in the `zeeguu/api` repository.
[4]The multi-level simplification prompt, `get_adaptive_simplification_prompt` in the `zeeguu/api` repository.

3.1.5 Solution. At its core, this is `map` for LLM calls: apply one shared, expensive prompt across a batch so its setup is paid once, not once per item. It takes two forms:

- ***Fan-in* batching** packs many independent inputs into one prompt, spreading a large instructional preamble across the whole batch.
- ***Fan-out* batching** produces many outputs from a single input, emitting one section per output and collapsing several requests into one.

Both combine naturally with *Anticipatory Precomputation*: because results are computed offline, there is the luxury of batching.

3.1.6 Consequences.

- **Cost and latency amortize with batch size.** The fixed preamble is paid once per call instead of once per item, so per-item token cost *and* wall-clock latency fall roughly inversely with the batch size.
- **Batch size is capped by the tightest ceiling, and which ceiling binds flips by direction.** For fan-in (many small items) the *quality* ceiling binds first (~15–20 items before the model drops or muddles entries), far below what the token window allows; for fan-out (full-length outputs) the *token* ceiling binds first, on the output side, since each result is full-length. So the workable batch size is workload-specific and must be tuned, not maximized.
- **Applies only to deferrable work.** Fan-in must wait to accumulate items, so the pattern fits offline / precomputed paths (composes with *Anticipatory Precomputation*) and is unavailable when a user is blocked on a single result.

3.1.7 Notes.

- **Both directions are the same `map`, over a different axis.** Fan-in maps the prompt over inputs, e.g. `map(validate, examples)`. Fan-out maps over outputs for a fixed input, e.g. `map(level $\rightarrow$ simplify(article, level), levels)`. Either way the shared prompt is the function whose fixed setup the batch pays for once.
- *Prompt caching is a partial substitute.* Some providers (e.g. DeepSeek) cache a repeated prompt prefix and discount it. That amortizes the preamble's *cost*, but not its *latency* or the per-call overhead of many round-trips, so batching still earns its keep even where the provider caches prompts.

3.1.8 Known Uses.

- **Batch prompting [3]** (Cheng, Kasai & Yu, EMNLP 2023) packs multiple independent samples under one shared instructional prompt in a single call, cutting token and time cost roughly inverse-linearly with batch size.
- The *fan-out* direction maps directly to structured-output calls that emit several keyed results at once, and to the OpenAI/Anthropic multi-output `n` parameter.
- *Distinguish from provider batch APIs.* The OpenAI Batch API and Anthropic Message Batches give ~50% off large asynchronous jobs, but each request still carries and pays for its own full prompt; they amortize scheduling and rate-limit overhead, *not* the in-prompt instructional overhead this pattern targets.

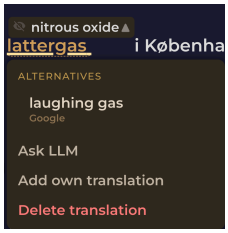


Fig. 2. In Zeeguu, the inline Google translation is the primary path; when the user wants a better rendering they escalate to an LLM on demand via the "Ask AI" option.

3.2 Escalate to the LLM

3.2.1 Context. A feature can be produced two ways: a cheap, fast specialized tool (a translation API, a classical classifier) that is adequate for the common case, and a slower, costlier LLM that does better on the hard cases. Crucially, the cheap tool's shortfalls are *observable*: it either errors, or the user visibly rejects the result.

3.2.2 Example. In Zeeguu, translation APIs (from Google, Azure, and DeepL) serve as the primary translation engines. When a user indicates the translation is inadequate (by choosing *Ask AI* from the alternatives menu), the system escalates to an LLM for a more nuanced, context-aware translation. This keeps costs low and speed high in the common case while providing higher, LLM-quality results when needed.

3.2.3 Problem. LLM quality is only worth its cost on the minority of requests the cheap tool handles poorly, and those cannot be told apart up front. How can the LLM be spent *only* where it pays off?

3.2.4 Forces. Specialized tools (translation APIs, NLP pipelines, classical classifiers) are faster, cheaper, and more deterministic than LLMs for well-defined tasks, but they sometimes fail or produce insufficiently satisfactory results.

3.2.5 Solution. Use the specialized tool as the primary path and escalate to the LLM only when the primary fails or the user signals dissatisfaction. The LLM receives the original input, and (where it helps) the specialized tool's output and the user's feedback alongside it, so the escalation refines rather than restarts.

3.2.6 Consequences.

- **Cheap in the common case.** The LLM runs only on failure or dissatisfaction, so the vast majority of requests never incur its cost or latency.
- **User-triggered escalation costs usability.** Unless it fires automatically on a detectable failure, the user must be given, and must notice and use, a way to signal dissatisfaction: extra UI surface, and the routing burden falls on them.
- **A miss doubles the wait.** When escalation happens the user has already waited for the cheap result first, so time-to-answer for those cases is the sum of both.

3.2.7 Notes.

- *Applies broadly.* Beyond translation: topic classification, named entity recognition, or any task where a cheaper tool handles the common case and the LLM handles the long tail.

- *Escalation, not fallback.* Unlike a reliability fallback (where the secondary is an equal-or-lesser backup invoked when the primary fails; see *Fail-Fast Provider Chain*), here the secondary is **more capable and more expensive**, invoked when the primary is not good enough. The movement is *up* in quality and cost, not *down* into degraded mode. That is why we name it escalation.
- *Relationship to the model cascade.* This is the human-/failure-triggered cousin of the **model cascade** in ML serving, where a cheap model runs first and a confidence threshold routes hard inputs to a larger model. The shared shape is *cheap tier first, expensive tier on demand*; the difference is the trigger. A cascade escalates automatically on the model's own low confidence, whereas this pattern escalates on external signals: the primary tool erroring, or the user explicitly declaring the result inadequate. A confidence-based cascade is thus one possible escalation policy; user dissatisfaction is another, and the two can be combined.

3.2.8 Known Uses.

- **FrugalGPT [2]** (Chen, Zaharia & Zou, 2023) queries cheaper models first and escalates to more capable, expensive ones only when a scorer rejects the cheap answer.
- **RouteLLM [9]** (Ong et al., 2024) trains a router that sends easy queries to a weak/cheap model and escalates only hard ones to the strong model; shipped as an open-source framework.
- These are the *model-cascade* cousins of the pattern (automatic, confidence-triggered) whereas Zeeguu's trigger is a failure of the cheaper *non-LLM* tool or explicit user dissatisfaction.

3.3 Slow-Path Inference

3.3.1 Context. A user-facing app makes many LLM calls, but only some are on a user's critical path. The rest (precomputation, batch classification, offline validation) are latency-insensitive. Latency-tolerant execution costs far less: an asynchronous batch endpoint runs the *same* model at roughly half price, and a cheaper provider or owned hardware less still.

3.3.2 Example. Zeeguu runs the pattern in two places.

Two model tiers for one task. Article simplification takes two paths, chosen by whether a reader is waiting.

1. When a learner opens an article that has not been simplified yet, the on-demand simplification runs on a fast model (Anthropic Haiku, the "real-time text-simplification" path), so the reader is waiting as little as possible.
2. The bulk, ahead-of-time simplification of the crawled article feed, which no user is blocking on, is pinned to the cheaper, slower DeepSeek model (`crawl_round_robin(..., simplification_provider='deepseek')`); the CEFR (reading-difficulty) assessment done during crawling is likewise DeepSeek-only, "for consistency with batch crawling".

Same task, two model tiers, selected by latency-sensitivity rather than by the task itself.

An owned machine as the slow path. Zeeguu's latency-insensitive LLM work (CEFR assessment, translation validation, meaning-frequency classification, ahead-of-time example generation) all bills against metered APIs, yet none of it is on a user's critical path. A single owned machine (a Mac Studio running Ollama) drains that entire queue overnight on a local model at zero per-token cost, with the cloud API as a deadline-bound fallback for whatever is not done by morning. It moves a whole category of spend off the metered bill onto hardware that already sits idle at night.

3.3.3 Problem. How can the premium cost of the real-time model be avoided on the large share of LLM work that no user is waiting for?

3.3.4 Forces.

- In a user-facing app, a large share of LLM work is *not* on a user's critical path: content pre-computed for later, batch classification, offline validation. Only a fraction is real-time.
- The real-time path must pay for low latency: a fast, often premium model. Paying those same rates for work no one is waiting on is wasteful.
- Latency-tolerant execution (a batch endpoint, a cheaper provider, owned hardware) is far cheaper than a premium real-time model, and fine for work no user is waiting on as long as its output is still good enough for the job. On the real-time path it is unacceptable.
- A single model for everything forces a bad compromise: premium cost on all of it, or premium latency and quality on none.

3.3.5 Solution. Split inference into a **fast path** and a **slow path**, and route each task by whether a user is blocking on the result, not by what the task is.

- **Fast path**: real-time, user-facing work runs on a low-latency model.
- **Slow path**: deferrable work (pre-computation, batch, offline) runs on the cheapest adequate path.

Realize the slow path with whatever is cheapest for the work, in rough order of adoption cost:

- **A cheaper model.** The lightest realization: point the batch path at a cheaper provider. Zeeguu uses DeepSeek for batch simplification and crawl-time CEFR assessment while the reader-facing path stays on Anthropic Haiku: the same call behind a different endpoint.
- **A provider batch API.** Asynchronous endpoints (OpenAI Batch, Anthropic Message Batches) run at roughly half price for up to ~24h turnaround, with no infrastructure to own.
- **Self-hosted local inference.** A local model on owned hardware draining an overnight queue at zero per-token cost: the largest saving, the most infrastructure (the owned-machine example above).

3.3.6 Consequences. Slow-path output is much cheaper and best-effort in timing, so only latency-insensitive work is eligible, and the fast path (or cloud) stays as a deadline-bound fallback (composes with *Fail-Fast Provider Chain* and *Escalate to the LLM*). Route only work whose slow-path result will actually be accepted: output that is rejected and recomputed on the fast path costs *more* than never taking the slow path at all. Results carry a different model identity and should be recorded as such (composes with *LLM Output Provenance*); untrusted slow-path output may need validation (composes with *LLM Content Validation Tracking*). Which model runs on which path should live in one place (composes with *Centralized Model Selection*), so re-tiering a feature is a one-line change.

3.3.7 Notes.

- *The owned machine is a pull worker.* It has no public IP and is not on the server's network, so it connects outbound-only: the server enqueues jobs, the machine polls over HTTPS and posts results back. A deployment detail, not an LLM concern.
- *Route by latency-sensitivity, not by task.* The same task, article simplification, runs on both paths; what selects the path is whether a human is blocked, not what is being computed. That is what separates this pattern from simply using a cheap model for cheap tasks.

3.3.8 Known Uses. This pattern is the LLM specialization of the long-standing **batch vs. online inference** split in production ML serving: an offline, high-throughput, latency-tolerant tier running alongside a real-time serving tier.

That split is the established instance-class; what is specific here is tiering the *same* task across cheaper and premium *LLM* endpoints by whether a user is waiting.

- **Apple Intelligence + Private Cloud Compute** is a shipped tiered-inference system (an on-device model for immediate requests, a heavier model on fallback), though it tiers by *capability* rather than latency-tolerance, and inverts the economics (local is the fast path).
- *Enablers, not instances.* The mechanisms that make a cheap slow path practical are widely used, but each supplies only the slow path, not the tiering *decision*: provider batch APIs at ~half price (OpenAI Batch, Anthropic Message Batches), local runtimes on owned hardware (Ollama), and outbound-only workers (GitHub Actions self-hosted runners). We did not find a documented LLM application that names this same-task, latency-tiered split: Zeeguu is our instance, and the batch/online ML practice its lineage.

3.4 Per-User Consumption Budget

3.4.1 Context. Some LLM-backed features are triggered directly by a user's action and bill a shared provider account the instant the user acts. Consumption is therefore driven by user behaviour, unbounded from the system's side, and can be extremely uneven across users.

3.4.2 Example. Several of Zeeguu's LLM actions are triggered *directly by a user's click* and charge a shared provider account the instant the user acts: on-demand "Ask AI" translation, on-demand article simplification, and audio-lesson generation (LLM script + text-to-speech). Because the spend is attached to individual user behaviour, a single enthusiastic user (or a buggy client, or a script) can run up the bill or exhaust shared rate limits for everyone.

Of all the above, the one that Zeeguu protects against is multiple lesson generations in parallel, because this is the most expensive operation. Audio-lesson generation allows **only one active generation per user at a time**: `AudioLessonGenerationProgress.create_for_user` deletes any existing record and `find_active_for_user` gates new requests. That is a per-user *concurrency* budget of exactly one, the cheapest possible bound, chosen precisely because the audio pipeline is among the most expensive actions in the system. On top of that, each user can generate only one audio lesson per day, a second, count-based budget.

The general pattern extends this idea to any LLM-backed resource: cap each user's consumption over a time window, denominated in whatever proxy is cheap to measure and *good enough*: number of on-demand translations per day, characters simplified per week, generations per hour, or, at the precise end, actual token cost.

3.4.3 Problem. How can one user, or a buggy client, be stopped from running up the shared bill or exhausting shared rate limits for everyone?

3.4.4 Forces.

- On-demand LLM actions cost real money and latency per invocation, and, unlike a pre-computed or cached feature, they are driven by user behaviour, so from the system's side consumption is unbounded.
- Consumption can be wildly uneven across users. A small number of heavy users, an abusive script, or a runaway client can dominate cost and exhaust the shared provider's rate limits, degrading service for everyone.
- The bound must be per *user* (the entity the spend is attached to), not global, or one user's overuse silently taxes the rest.

- Exact cost accounting is precise but comparatively expensive to build (capture token usage, maintain a current price table). Often a coarse proxy (a count, a concurrency cap, a time budget) is far cheaper and adequately bounds the failure that actually matters.

3.4.5 Solution. Give each user a budget on an LLM-backed resource and refuse or degrade once it is exhausted. Denominate the budget in the **coarsest proxy that adequately tracks the risk**:

- **concurrency**: at most N in flight per user (the audio-lesson case, $N = 1$);
- **count over a window**: translations per day, simplifications per week;
- **volume**: characters simplified, tokens consumed;
- **actual cost**: per-user cost attribution, the precise variant (see below).

When a budget is exhausted, **degrade rather than fail** where possible: fall back to the cheaper non-LLM path (serve the Google Translate result and simply stop escalating to the LLM; see *Escalate to the LLM*), queue the request, or show a friendly "try again later."

The precise variant, per-user cost attribution. At the accurate end, meter every LLM call as `(user, feature, model, provider, input_tokens, output_tokens, timestamp)`, store **tokens** (provider-neutral) and derive **cost** through a central price table, and aggregate per user. This is the most accurate budget denomination and doubles as observability: it answers *which users and which features drive the bill*. It also composes with *Centralized Model Selection* (price table keyed by the same model constants) and *LLM Output Provenance* (the per-artifact provenance record is the natural place to also stamp token counts: provenance says *how was it made*, this adds *what did it cost, and to whom*). Prefer a coarse proxy unless this precision is genuinely needed.

3.4.6 Consequences.

- **One user can no longer tax the rest.** A heavy user, a runaway client, or an abusive script is bounded to its own budget, so it cannot dominate cost or starve everyone else's share of the provider's rate limits.
- **The budget is a thing to size and maintain.** It needs a denomination and a window, and the choice trades effort against tightness: a coarse proxy (a count, a concurrency cap) is cheap to build but loose, while per-token cost accounting is precise and the most work.
- **What happens at the limit is a product decision, not a mechanism.** Degrading to a cheaper non-LLM path keeps the feature friendly (composes with *Escalate to the LLM*); a hard refuse does not. No gateway can make that call.

3.4.7 Notes.

- Choose the coarsest proxy that prevents the failure in question. If the risk is "a script hammers *simplify*," a per-hour count stops it; per-token accounting is not needed.
- Attribute to the provider that **actually** served the call: a *Fail-Fast Provider Chain* fallback changes which provider is billed and at what rate.
- The graceful-degradation clause is what keeps this user-friendly rather than punitive: a language learner who exhausts their LLM-translation budget still gets Google Translate, not an error.

3.4.8 Relationship to LLM Gateways. Gateways provide per-key / per-user rate limiting and spend caps out of the box (LiteLLM virtual-key budgets and RPM/TPM limits, Portkey, Cloudflare AI Gateway), so both the coarse and the cost-based variants can be *partly* delegated, but only if the application tags each call with a user id (and, for per-feature

budgets, with the feature). As with metering generally (see *What Makes These Patterns LLM-Specific? → Relationship to LLM Gateways*), the gateway supplies the **enforcement primitive** (count, throttle, cap); the pattern owns the **policy**: which resource, which window, and crucially *what the user gets when the budget is exhausted* (degrade to Google Translate vs. hard refuse), which is a product decision no gateway can make.

3.4.9 Status. A crude instance (single active audio generation per user) already ships. The general per-user budget across on-demand actions, and the cost-accounting variant, are not yet implemented, included as a candidate to invite discussion on whether this is one pattern with several denominations (concurrency / count / volume / cost) or several distinct patterns.

3.4.10 Known Uses.

- **GitHub Copilot** gives each user a monthly budget of "premium requests"; when it is exhausted the user is not blocked: "you can still use Copilot with one of the included models for the rest of the month" (subject to rate limiting). A per-user count budget that degrades to a cheaper model tier.
- **Canva** caps per-user AI usage by plan; paid users who hit the limit see "short pauses between generations" (a throttle) rather than a hard stop.
- **Cursor** (historical, pre-2025) gave each user 500 "fast requests"/month, then degraded to a slower unprioritized queue rather than blocking.
- *Enablers.* Gateways offer per-user budgets as a primitive (LiteLLM, Portkey, Helicone): the enforcement mechanism, not the *policy* of what the user gets when exhausted.
- *Observed.* Public examples degrade to a cheaper model tier or a slower queue; degrading to a genuinely non-LLM path (as Zeeguu does with Google Translate) appears rarer in the wild.

4 Latency and Availability

LLM calls take seconds, and the providers behind them err, rate-limit, and spike without warning, yet users on the critical path expect an answer in well under a second. These patterns keep the slow, unreliable model off that path. *Anticipatory Precomputation* moves the work earlier, computing likely-needed results offline so they are ready before the user asks. *Hot-Path Result Caching* moves it sideways, serving a repeated result from memory instead of recomputing it. *Multiplexed Dispatch* races several models or providers and takes the first response, so worst-case latency tracks the fastest rather than the slowest. *Fail-Fast Provider Chain* keeps the request alive when a provider dies, falling straight through to the next one instead of spending the latency budget on retries.

4.1 Anticipatory Precomputation

4.1.1 Context. A user-facing feature needs an LLM result, but the model takes seconds while the user expects a response in well under one, and *which* results a given user will need next is predictable from their behaviour.

4.1.2 Example. The vocabulary exercises a learner practices are built from the words they looked up while reading, each paired with a machine translation. At reading time an imperfect translation is low-stakes: the reader is only making sense of the text, and can pick a better option from the alternatives menu. But once the platform selects a word for the learner to *learn*, they will drill that word-translation pair over many days, so its correctness suddenly matters. A regular cron job looks ahead: it finds the words each learner is due to study next and validates them with an LLM, so

that when the exercise module asks for new words, the vetted ones are ready straight away. If none are precomputed yet (the learner translated a few words and jumped straight into exercises), the validation runs in real time.

An even costlier instance: audio lessons. Generating a personalized audio lesson is more expensive again: an LLM writes the lesson script, then text-to-speech synthesizes the audio, several seconds of work no learner should wait through. A nightly job pre-computes the next lessons for recently active learners (prioritized by how recently they practiced), on the assumption that someone who has been studying will be back for the next one. When they return, the lesson is already waiting.

4.1.3 Problem. How can an LLM-quality result reach the user's critical path without a wait for the model, when upcoming needs are often predictable?

4.1.4 Forces. LLMs can provide valuable data for users, but they are slow and expensive, making their invocation impractical when the user needs an answer in real-time. (Real-time users expect answers in 200ms, while depending on the prompt and the deployment configuration, an LLM-based system can take multiple seconds to produce an answer).

4.1.5 Solution. Anticipate likely user needs and pre-compute LLM results offline (e.g., via cron jobs), so results are available instantly when needed. The system designer should model user behavior in order to predict their LLM needs.

4.1.6 Consequences.

- **Zero latency at request time.** The LLM's cost and multi-second latency are paid entirely off the hot path; when the user acts, the result is already waiting.
- **Precomputing pays for guesses that miss.** It spends tokens on results that may never be requested, so the value depends on the accuracy of the behaviour model: a poor predictor wastes spend *and* still misses.
- **Needs a reliable "what" and "when" signal, plus a fallback.** It applies only where upcoming needs are predictable; cold or mispredicted requests still need an on-demand path. Composes with *Prompt Amortization* (offline results can be batched).

4.1.7 Known Uses.

- **Yelp** pre-computes LLM query-understanding responses for high-frequency ("head") search queries into a key/value store, "caching (pre-computing) high-end LLM responses for only head queries", reaching 95% of traffic for review-highlight expansions, so the expensive model never runs on the hot path.
- **Instacart**'s Intent Engine serves high-frequency search queries from a precomputed cache of LLM outputs, leaving only ~2% of queries to a real-time model.
- **ColBERT [6]** (Khattab & Zaharia, SIGIR 2020) precomputes contextualized passage embeddings for the whole corpus offline, so query time runs only cheap late-interaction matching: the canonical "precompute the expensive representation ahead of time."
- *Enabler, not instance.* Provider batch / offline-inference endpoints are the infrastructure for computing LLM outputs in bulk off the hot path, not a use of the pattern.

4.2 Hot-Path Result Caching

4.2.1 Context. The same or near-identical LLM request recurs within a short window (many users hitting the same content, or one user re-requesting) but *which* requests recur is unpredictable, so they cannot be precomputed ahead of time.

4.2.2 Example. Any expensive LLM call whose inputs recur can be cached in memory, keyed on those inputs, so every repeat after the first is near-free. Zeeguu does this for Multi-word expression (MWE) detection, which finds the phrases in an article a learner might want translated as a unit (for example *break the ice*). It runs an LLM, gated by a cheap Stanza (an NLP library) pass (see *Hybrid Classical+LLM Pipeline*), and the LLM call is the expensive part. The detector keeps a 500-entry in-memory cache keyed on the article's sentences (dropping its oldest entries when full), so when many users read the same article, the analysis computed for the first reader is served instantly to the rest. Hit rates are highest for popular articles read by many learners in the same window. Only LLM results are cached: when the LLM is unavailable and the detector falls back to Stanza, those weaker results are deliberately not stored.

4.2.3 Problem. How can the same expensive LLM call be avoided when it recurs unpredictably within a short window?

4.2.4 Forces. Pre-computation handles predictable needs, but some LLM queries are repeated unpredictably within short time windows (e.g., multiple users encountering the same phrase, or a single user re-requesting the same analysis). These don't justify persistent storage but do benefit from short-term caching.

4.2.5 Solution. Maintain an in-memory cache for recent LLM results, keyed on the call's inputs. Bound it by capacity and evict the oldest entries when full (the example above caps the cache at 500 entries). No time-based expiry is needed where the cached result is a pure function of its inputs.

4.2.6 Consequences.

- **Repeats are near-free.** A hit within the window returns from memory at ~zero cost and latency; hit rate is highest for popular content many users read in the same window.
- **Memory is the price, and it only pays where inputs repeat.** The cache costs memory and returns nothing on a wide, flat input space; it fits where requests cluster on the same content, so the same calls actually recur.
- **Variant: persist it.** A DB-backed cache outlives process memory and survives restarts (Zeeguu uses this too), trading a little speed for durability.

4.2.7 Known Uses.

- **Walmart**: Walmart's chief software architect describes a production *semantic* cache (vector-similarity, not exact match) for e-commerce search, reporting a hit rate "closer to 50%" on tail queries. *(Reported in a vendor writeup of the talk; the claims are attributed to Walmart on the record, but the source is secondary.)*
- *Tools that ship this, not documented deployments.* Result caching is common enough to be productized: dedicated LLM caches (GPTCache [1], exact + semantic) and gateway caches (Helicone, Portkey, LangChain).
- *Honest note.* First-party write-ups of *reactive* LLM-output caching are scarce; most published examples (Yelp, Instacart) are *anticipatory precomputation* of head queries (see *Anticipatory Precomputation*) rather than caching on a miss.

4.3 Fail-Fast Provider Chain

4.3.1 Context. A request is served by one of several interchangeable LLM providers, on the user's critical path, and any provider can intermittently error, rate-limit, or spike in latency.

4.3.2 Example. (Zeeguu): The app uses a unified LLM service proxy which tries Anthropic first; on any error (timeout, rate limit, API error), it immediately switches to DeepSeek without retry. This keeps worst-case latency bounded while maintaining availability.

4.3.3 Problem. How can the request keep succeeding within its latency budget when a provider fails, without burning that budget on retries?

4.3.4 Forces. LLM providers experience outages, rate limits, and latency spikes. Retry logic increases latency and may still fail. Different providers offer similar capabilities at different price/performance points.

4.3.5 Solution. Configure a chain of LLM providers with no retries. On any failure, immediately fall back to the next provider in the chain. Prioritize speed over exhausting retry budgets.

4.3.6 Consequences.

- **The request survives an outage, and latency stays bounded.** A dead or slow provider no longer fails the request, and skipping retries keeps the worst case tight.
- **It shines on real-time paths, less so offline.** Where a user waits, spending the budget on a retry against a failing provider is the wrong move. For offline or batched work nobody is waiting, so retry-and-backoff, even retrying the cheaper provider first, is more affordable.
- **Cost attribution shifts to whoever served.** A fallback changes which provider is billed and at what rate (composes with *Per-User Consumption Budget* and *Centralized Model Selection*).

4.3.7 Note. This differs from *Escalate to the LLM* in that all components in the chain are LLMs offering equivalent capabilities: this is a *fallback* for reliability, not the *escalation* to a more capable (and more expensive) tier.

4.3.8 Known Uses.

- **Karrot** (Korea's largest secondhand marketplace) runs a tiered resilience chain on its GenAI platform: retry, then region failover, then cross-provider fallback: "If Google's `gemini-2.5-flash` is unavailable even across regions, we route to OpenAI's GPT-5."
- **Slack**'s routing layer trips a circuit breaker on latency/error spikes and diverts to a designated backup: "we always designate backup models for every feature; if the primary choice doesn't meet our performance or quality thresholds in real-time, the system knows exactly where to go next."
- **Digits** routes LLM calls through an internal proxy specifically to fail over between providers, because "neither OpenAI nor Anthropic maintain 100% uptime."
- **Whatnot** treats reliability as a platform concern, adding "multi-provider support, moving toward fallback behavior by default."
- *Tools that ship this.* Gateways provide fallback chains as a configurable feature (Portkey, Cloudflare AI Gateway, LiteLLM Router, which defaults to retry-then-fallback), the enforcement mechanism, not a deployment.

4.4 Multiplexed Dispatch

4.4.1 Context. Several models or providers can produce the same result at different and individually *variable* latencies, and the request sits on the user's critical path where a slow tail hurts the experience.

4.4.2 Example. Zeeguu races calls on its real-time paths, where a waiting user feels every slow response.

- **On-demand simplification.** When a reader opens an article that has to be simplified right now, the request goes to two models at once; the first response is rendered, and the slower one is kept as an alternative the reader can switch to. Racing collapses the latency tail on the one path where a human is blocked, and it doubles as failover: if one provider is slow or down, the other still answers. The redundant cost is affordable because on-demand simplification is rare next to the crawled batch.
- **Translation.** Real-time translations are likewise dispatched to several providers in parallel; the first response is used and the rest are surfaced as alternative translations.

4.4.3 Problem. How can worst-case latency stay low when any single provider can be intermittently slow?

4.4.4 Forces. Multiple LLM providers offer similar capabilities but with varying latency. When speed matters for user experience, relying on a single provider creates a bottleneck.

4.4.5 Solution. Dispatch the same request to multiple providers simultaneously and use the first response that arrives. To cap the redundant cost, race only the few providers that have historically been fastest rather than all of them.

This composes with **live retrieval**: when a user is unsure of a translation and asks for alternatives, the UI shows the already-raced cached results immediately, then fetches more on demand, streaming them in as they arrive.

4.4.6 Consequences.

- **Latency tracks the fastest responder, not the average.** Racing collapses the slow tail: worst-case latency becomes the *best* of N providers rather than any one's.
- **N× the cost for one used result**: every dispatch bills its own tokens. (Zeeguu recoups some of this by keeping the losing responses as alternatives; see the note below.)
- **It doubles as failover.** Each racer is a full, independent attempt, so a provider that errors or hangs simply loses the race instead of failing the request: availability comes for free, where *Fail-Fast Provider Chain* buys it with an explicit fallback.

4.4.7 Note. In itself this is not LLM-specific. It is the *hedged requests* pattern from distributed systems (Dean and Barroso, *The Tail at Scale*, CACM 2013), also known as request racing or tied requests. Two things give it an LLM flavour here. 1. First, the hedge is across independent, competing vendors (Anthropic, DeepSeek, Google Translate) rather than replicas of a single service. 2. Second, each redundant call costs real money per token, so the losing responses are not thrown away: they are kept and surfaced to future readers of the same text as alternative translations, turning the redundant work into a feature.

4.4.8 Known Uses.

- **Hedged requests [4]** (Dean & Barroso, "The Tail at Scale," CACM 2013) are the classical ancestor: send a duplicate to another replica after a latency threshold, take the first, cancel the rest, cutting BigTable p99 from 1800ms to 74ms at ~2% extra load.

- General infrastructure ships it: **gRPC `hedgingPolicy`** and **Polly** (.NET) race concurrent copies and take the first response.
- *Honest gap / opportunity.* We could not find a major LLM gateway that ships true provider *racing* as a first-class feature; most do sequential fallback (see *Fail-Fast Provider Chain*) or learned single-model routing. Zeeguu's parallel dispatch, across translation providers and across two models for on-demand simplification, appears to be an under-adopted transfer of the hedging idea to LLMs.

5 Lifecycle Management

An LLM integration is not static: the models behind it are retired and re-priced on the vendor's schedule, and the role the LLM plays in a feature changes as the system matures. These patterns manage that evolution over time. *Rent, Then Build* treats the LLM as a temporary stand-in, shipping a feature on rented general capability now while a cheaper, dedicated replacement is built. *Centralized Model Selection* keeps every model identifier in one place, so a vendor retiring a dated snapshot is a one-line change rather than a hunt across the codebase.

5.1 Rent, Then Build

5.1.1 Context. A feature needs to work in production now, but the efficient, dedicated version of it (a fine-tuned or classical model) needs training data or engineering that does not exist yet. A general-purpose LLM can do the task immediately, if expensively.

5.1.2 Example. Topic classification of articles is currently performed by pasting the abstract into an LLM and asking it to classify the topic. This works but is expensive. Once we are satisfied with the topic taxonomy and have accumulated enough labeled data, we will switch to a dedicated topic detection framework.

5.1.3 Problem. How can a feature ship and start earning its keep before the cheaper, dedicated version that will eventually run it has been built?

5.1.4 Forces. LLMs are general-purpose machines that can attempt almost any text-based task. Building dedicated, efficient solutions requires upfront investment and training data that may not yet exist.

5.1.5 Solution. Use the LLM to perform a task in production while building a more efficient replacement. The arc is rent, then build: the LLM's general capability is *rented* (paid per call, costly but available immediately) to ship the feature now, while a cheaper, dedicated implementation is *built* to eventually own the task. The rented LLM is convincing to users and good for early beta-testing and feedback, but intended to be temporary.

Bootstrapping variant: In a more subtle variant, the LLM *generates the training data for its own replacement.* For example, Zeeguu uses an LLM to estimate text difficulty levels. These LLM-generated difficulty labels are being accumulated as training data for a classical classifier that will eventually take over the task.

5.1.6 Consequences.

- **The feature is live from day one.** It gathers real usage and feedback immediately, with the LLM standing in for a component that does not exist yet.
- **Running the stand-in funds its successor.** The LLM's inputs and outputs accumulate as the labeled data the dedicated replacement needs, so the temporary solution also produces the training set for the permanent one (the bootstrapping variant).

- **The stand-in is expensive, and "temporary" can stick.** Until the replacement ships, the feature runs at LLM cost and latency, the replacement is a second system to build and validate, and the intended-temporary LLM can quietly become permanent.

5.1.7 Note. This pattern has a lifecycle relationship with *Escalate to the LLM*: a system may start with the LLM as primary (renting it), migrate to a specialized tool as primary, and then keep the LLM as the escalation path, completing a full cycle from LLM-first to LLM-on-demand.

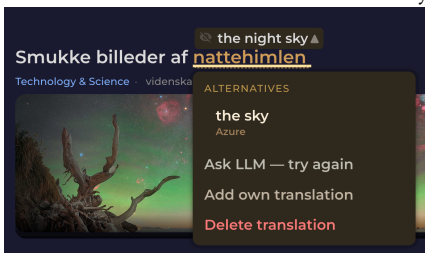
5.1.8 Known Uses.

- **OpenAI Model Distillation** (Stored Completions) captures a large model's production input–output pairs and fine-tunes a smaller model as a drop-in replacement: the pattern shipped as a product feature.
- **Self-Instruct / Alpaca [12]** (Wang et al.; Taori et al., 2023) use a strong LLM to generate the instruction data that fine-tunes a small replacement: the bootstrapping variant.
- **Distilling Step-by-Step [5]** (Hsieh et al., ACL Findings 2023) trains task-specific models up to 700× smaller from LLM-generated rationales.

5.2 Centralized Model Selection

5.2.1 Context. A single LLM model identifier is referenced from many call sites across a codebase. Providers retire dated model snapshots on their own schedule, so an ID that is valid today can start returning 404 at a future date with no change to the code.

5.2.2 Example. Zeeguu's reader has an on-demand "Ask AI" translation action. Its model ID, `claude-sonnet-4-20250514`, was hardcoded at three call sites in the translation service and two more in the multi-word-expression (MWE) detector. Anthropic retired that snapshot; every call began returning `404 not_found_error`. The error was swallowed one layer down (the helper returns `None` on any exception), so the endpoint returned a generic `404 "LLM translation failed"` and the reader silently degraded to an "**Ask AI, try again**" button that could never succeed.



Nothing in the code had changed; a date had passed. The error started showing up.

The fix itself was mechanical: swap to the live snapshot the rest of the codebase already used. But finding it took a dig through production logs, and the swap had to be repeated at five sites.

To avoid this happening again in the future, Zeeguu now keeps every model identifier in one module, keyed by *role*, the job a model does in a feature (`WORD_TRANSLATION`, `MWE_DETECTION`, `SIMPLIFICATION`, …), and each resolving to a canonical vendor ID declared once; a feature asks for `models.WORD_TRANSLATION` and never names a snapshot, so the next retirement is a one-line edit.

5.2.3 Problem. How can a provider's model retirement be survived without hunting down every hardcoded model ID scattered across the codebase?

5.2.4 Forces.

- Model identifiers are volatile in a way ordinary configuration is not
- Providers deprecate and retire dated snapshots on *their* schedule, so a hardcoded ID that works today returns a 404 at some future date, with no change to the code at all.
- Model choice is also cross-cutting: the same handful of IDs get copy-pasted across many features and several cost/quality tiers, so a single retirement breaks many call sites at once, and there is no one place to see, or change, which model each feature uses. Because the failure is a runtime error on a string that looks perfectly valid, it survives code review and type-checking.

5.2.5 Solution. Keep every model identifier in one central module. Declare the canonical vendor IDs once, then expose *role-based aliases* (one per use: translation, classification, simplification…) that resolve to them. Call sites import the role, never the raw string. Surviving a vendor retirement, or moving one feature to a cheaper or faster tier, becomes a one-line change in a file that documents the whole model landscape on one screen.

5.2.6 Consequences.

- **A retirement is a one-line edit.** Surviving a deprecation, or re-pointing a feature to a cheaper or faster tier, changes one line in a module that shows the whole model landscape on one screen, rather than a hunt across call sites.
- **The registry only helps if nothing bypasses it.** It adds one layer of indirection, and a single stray hardcoded ID reintroduces the exact failure the pattern prevents, so the discipline has to hold everywhere.
- **One constant keeps selection and provenance in sync.** Because the same central constant both selects the model and stamps it onto output, the two cannot drift (composes with *LLM Output Provenance*); it also names which model each entry of a *Fail-Fast Provider Chain* uses.

5.2.7 Notes.

- The mechanism is classical (single source of truth; no magic strings). What makes it an LLM pattern is the *force*: vendor-driven model deprecation turns an innocuous hardcoded string into a scheduled runtime failure. It is the direct defense against the *"old ones regularly deprecated"* property listed among the core LLM forces in the introduction.
- Prefer **role-based** aliases over vendor-based constants. `WORD_TRANSLATION = HAIKU` reads as intent and lets one re-point a single feature's tier without touching others; a bare `HAIKU = "claude-haiku-…"` re-exported everywhere quietly couples unrelated features to the same choice.
- *Alternative: AI gateway.* An AI gateway can host the role→model mapping as named aliases in its config, relocating this pattern's registry out of application code. See the broader treatment of what gateways do and do not subsume in *What Makes These Patterns LLM-Specific? → Relationship to LLM Gateways.*

5.2.8 War Story. The same commit that repaired the retirement uncovered a second casualty of the identical disease. Simplified articles were being stamped `ai_model="claude-3-5-sonnet"` while the simplifier had long since moved to Haiku. The provenance field named a model that was no longer even in the pipeline (see *LLM Output Provenance*). Same root cause as the retirement: a model identifier duplicated into a place nobody remembers to update, silently

going stale. Centralizing selection lets the provenance stamp and the call-time choice read from the same constant, so they cannot disagree.

5.2.9 Known Uses.

- **Sourcegraph Cody** binds each feature *role* (`chat`, `fastChat`, `codeCompletion`) to a `modelRef` in a central `defaultModels` map, so changing which model powers a feature is a single config edit rather than a call-site change: genuine role-based central selection in a shipped product.
- *Enablers.* Gateways relocate the role→model map into their config as aliases (LiteLLM model aliases, Portkey Model Catalog): the same mechanism, hosted outside app code.
- The mechanism is the classic *single source of truth*; what makes it LLM-specific is vendor-driven model deprecation (see this pattern's forces). We did not find a public first-hand account of centralizing model IDs *in response to a deprecation outage*: Zeeguu's is our instance.

6 Data Management

LLM output that lands in persistent storage becomes a long-lived asset, reused long after the prompt and model that produced it have moved on. These patterns manage that stored output as it ages. *LLM Output Provenance* stamps each artifact with the model and prompt that made it, so exactly the stale ones can be found and regenerated when something improves. *Soft Invalidation of LLM Artifacts* then retires the stale ones without breaking the past, deprecating rather than deleting and regenerating lazily on next demand while old references keep resolving.

6.1 LLM Output Provenance

6.1.1 Context. LLM-generated artifacts (example sentences, summaries, labels) are written to persistent storage and reused for a long time, while the models and prompts that produce them keep improving. The prompt changes more often, and often matters more to output quality, than the model.

6.1.2 Example. When the system generates example sentences with a given word to be used in exercises, it stores which model and prompt version produced each result. When a prompt is improved, the system can identify and regenerate stale outputs without reprocessing everything.

6.1.3 Problem. When a prompt or model improves, how can exactly the stale artifacts be found and regenerated, without reprocessing the entire store?

6.1.4 Forces. LLM-generated data that enters persistent storage becomes a long-lived asset, but models and prompts improve over time. Without knowing how a piece of data was generated, it cannot be selectively regenerated when better models or prompts become available. Prompts evolve more frequently than model versions and can have a larger impact on output quality.

6.1.5 Solution. Store the full provenance tuple alongside every LLM-generated artifact: (model version, prompt version, generated output, timestamp). This enables selective regeneration (e.g., *"re-run everything produced by prompt v2 with the improved prompt v3"*) and quality auditing.

6.1.6 Consequences.

- **Selective regeneration becomes a query.** Re-run everything a given prompt or model produced and leave the rest, and the same record doubles as a quality-audit trail.
- **The stamp must be present and precise.** Every write has to record the provenance, and it is only as useful as the granularity it captures: a field that is not updated when the prompt is edited in place silently goes stale and drives nothing.
- **One identifier, shared with selection and validation.** The stamped model identifier and the one used to select the model at call time should be the same central constant (composes with *Centralized Model Selection*), and provenance pairs with *LLM Content Validation Tracking*: how an artifact was made, and whether it has been confirmed.

6.1.7 Notes.

- The key insight is that the prompt is at least as important to version as the model: a prompt change can completely alter output format, quality, or behavior even with the same model.
- This is also critical for *Rent, Then Build*: when accumulating LLM-generated labels as training data for a classical replacement, provenance tracking lets one exclude data produced by a prompt version that was later found to be noisy or biased.
- A field that names a model no longer in the pipeline is worse than no field at all, so stamp the provenance from the same constant used to *select* the model (*Centralized Model Selection*).
- Implicit provenance: Keep model names and prompt versions as constants in code. When one needs to know what generated a piece of data, correlate its `created_at` timestamp with git history to determine which model/prompt was deployed at that time. However, this works for simpler systems where there is a single model/prompt active at any time. A system using alternative prompts, e.g. for A/B testing, will have to track provenance explicitly. Also, explicit tracking makes data analysis faster, and ensures that data is self-describable.

6.1.8 Known Uses.

- *Better attested in tooling than in production self-reports.* The capability is productized: MLflow Prompt Registry and LangSmith version prompts and record which prompt/model produced each output, confirming the pattern's core claim that the *prompt* deserves versioning as much as the model, but these are tools, not documented in-app deployments.
- *Adjacent production pipelines.* DoorDash stores versioned LLM-generated profiles and "treat[s] prompts as code"; Etsy stores Pydantic-validated LLM attribute extractions over 100M+ listings. Both store LLM artifacts at scale and version prompts, but neither publicly describes stamping *each artifact* with its prompt version to drive *selective* regeneration: the load-bearing mechanic here.
- We did not find a first-hand account of the full pattern; Zeeguu is our instance.

6.2 Soft Invalidation of LLM Artifacts

6.2.1 Context. An LLM-generated artifact has been stored and is reused as a cache, and it is also referenced from user-visible history. Then the prompt or model that produced it improves, so the stored version is now known to be suboptimal while old references still point at it.

6.2.2 Example. When the prompt that generates audio lesson scripts was improved, the ~900 stored `audio_lesson_meaning` rows (each caching one vocabulary item's generated audio) produced under the previous prompt were neither regenerated eagerly nor deleted. Instead, each affected row received a `deprecated_at` timestamp, and the cache-lookup helper (`AudioLessonMeaning.find()`) was gated to skip deprecated rows. New daily lessons request a fresh row and trigger regeneration under the new prompt; existing daily lessons that already reference a deprecated row keep playing their old audio without breaking.

6.2.3 Problem. When the generator improves, how can stored artifacts be refreshed without either paying to regenerate everything up front or breaking the past references that still point at the old ones?

6.2.4 Forces. When a prompt or model improves, the obvious responses each have a serious drawback: - *Regenerate everything eagerly*: expensive, floods generation queues if affected rows number in the thousands, and pays for content that may never be re-requested. - *Delete the stale rows*: breaks any downstream object that references them by id (history, analytics, user-visible past sessions). - *Leave the stale rows in place and accept future reuse*: silently propagates the old, known-suboptimal quality.

None of these are good defaults for production systems where LLM-generated artifacts are referenced from user-visible history and are also targets for reuse.

6.2.5 Solution. Mark stale rows as deprecated rather than mutating or removing them. Gate the cache-lookup / reuse path to skip deprecated rows, forcing fresh generation on next demand. Existing references to a deprecated row remain valid (the row keeps its content for historical playback), but no new consumer picks it up. Regeneration cost is paid lazily, amortized over normal access patterns, and only for content that is actually requested again.

6.2.6 Consequences.

- **Cost is lazy and history stays intact.** Regeneration is paid only for content requested again, and old references keep resolving to their original artifact, so user-visible history does not break.
- **Old versions linger until next demand.** A replay hears the old quality until something triggers regeneration, and the deprecation flag has to reach every downstream cache to be effective.
- **It assumes artifact identity follows the row.** If the stored artifact is keyed by its source (e.g. `meaning_id`) rather than its own row id, a regenerated row overwrites the deprecated one's file (see the note). Pairs with *LLM Output Provenance*: provenance says which rows are stale, this says what to do once that is known.

6.2.7 Notes.

- This pattern is forward-only: it gates *reuse*, not *playback*. A user replaying an old lesson hears the old (lower-quality) version. That is usually preferable to a silent content swap mid-history.
- Works best when content has a clear "next access triggers regeneration" entry point. If consumers cache aggressively further downstream, the deprecation flag has to propagate to those layers too.
- Composes naturally with *LLM Output Provenance*: provenance answers "which rows are stale?", soft invalidation answers "what do I do with them once I know?".
- **Prerequisite: artifact identity must follow the row, not the upstream key.** If the on-disk artifact (audio file, image, embedding) is named after the *source identity* (e.g. `meaning_id`) rather than the *row identity* (e.g. `audio_lesson_meaning.id`), the regenerated row's artifact overwrites the deprecated row's artifact on the same path, defeating the historical-playback guarantee. Zeeguu encountered this concretely: meaning-lesson audio

files were keyed by `meaning_id`, so regeneration silently replaced the audio referenced by old daily lessons. A separate change re-keyed those files by row id to make Soft Invalidation safe. This small structural requirement may deserve being a pattern in its own right (working title: *artifact identity = row identity*).

6.2.8 Known Uses.

- **`stale-while-revalidate`** (IETF RFC 5861) is the same mechanic in HTTP caching: keep a stale entry usable and revalidate it lazily on next demand rather than eagerly regenerating.
- The **soft-delete / tombstone** database idiom marks a row with a timestamp and gates every read (`WHERE deleted_at IS NULL`), so the row stays intact for existing references but drops out of the active path: structurally identical to a `deprecated_at` gate.
- *Analogues, not exact instances.* Both capture the mechanism, but we did not find a documented LLM system that combines mark-deprecated + gate-reuse + **keep-old-rows-resolvable-for-user-history** + lazy regeneration; the history-preservation facet appears novel, so we present it as our own contribution, grounded in Zeeguu.

7 Quality Assurance

LLMs are imprecise: they make mistakes, and their output is only probabilistically well-formed. These patterns guard the boundary between that output and the code, data, and users that expect it to be correct. *Hybrid Classical+LLM Pipeline* lets a cheap classical tool gate the LLM, running it only where its judgment is actually needed. *LLM-Checking-LLM* catches errors with a second, narrower verification call. *Defensive Output Parsing* refuses to trust the shape of a response, parsing in layers and degrading gracefully rather than crashing on a malformed one. *Deterministic Postprocessing* fixes stable, judgment-free defects in code rather than in the prompt. *LLM Content Validation Tracking* records how far each stored artifact has been trusted, so consumers can gate on it. *Targeted User Feedback* closes the loop, letting users flag the errors that slip through, anchored to the exact place they occur.

7.1 LLM-Checking-LLM

7.1.1 Context. An LLM generates content that will be used or stored, and its output is sometimes wrong in ways a targeted check could catch. Verifying a specific property (grammaticality, factual match, difficulty level) is a narrower task than the open-ended generation that produced it.

7.1.2 Example. After generating contextual example sentences for vocabulary words, a second LLM call reviews the examples for accuracy, naturalness, and appropriate difficulty level.

7.1.3 Problem. How can an unreliable generator's mistakes be caught, when a second generator would be just as unreliable?

7.1.4 Forces. LLMs are imprecise generators, but verification of specific properties (e.g., grammatical correctness) is a more constrained task than open-ended generation (e.g., article simplification, rewriting a text at a simpler reading level). A second, focused LLM call can catch errors that the first, more complex call introduced.

7.1.5 Solution. Use one LLM call to generate a result, then use a separate LLM call to check or refine it. This escapes the paradox because verifying one specific property (is it grammatical? does it match the source?) is a narrower task than the open-ended generation that produced the output, so the checker fails less often on that property than the generator did. The verification prompt can be simpler and more focused than the generation prompt.

7.1.6 Consequences.

- **A focused check is more reliable than the generation.** Verifying one property is easier than producing the whole output, so the second call catches errors the first introduced, for the price of one extra call.
- **It narrows the error rate, it does not remove it.** The checker is itself an LLM and can return its own false verdicts, and it adds cost and latency.
- **It pays off only when checking is genuinely narrower than generating.** A check as open-ended as the generation buys little. The verdict composes with *LLM Content Validation Tracking* (record it) and *Hybrid Classical+LLM Pipeline* (a classical check is cheaper still, where one exists).

7.1.7 Note. This is distinct from ensemble methods or chain-of-thought: the key insight is that checking is a fundamentally easier task than generating, and this asymmetry can be exploited architecturally.

7.1.8 Known Uses.

- **LLM-as-a-judge [13]** (Zheng et al., NeurIPS 2023) uses a separate strong LLM to score another model's open-ended outputs.
- **Self-Refine [8]** (Madaan et al., NeurIPS 2023): a separate pass critiques and refines the first output.
- **G-Eval [7]** (Liu et al., 2023), shipped in eval frameworks such as **DeepEval**, uses a separate chain-of-thought judge call to grade generated output.
- *Contrast.* Unlike self-critique on the same reasoning, this pattern uses a differently-prompted call to check a *different property* (see *Related Work* on Stechly et al.).

7.2 LLM Content Validation Tracking

7.2.1 Context. LLM-generated content is written into persistent storage alongside human-verified data, and downstream features and users will read it. Some of it has been checked, most has not, and once stored the two look identical.

7.2.2 Example. A word-translation pair can exist at several trust levels: generated by Google Translate (unverified), verified by an LLM checker (auto-verified), implicitly accepted (a user drilled it in an exercise without flagging it as wrong), or explicitly corrected by a user. Each level carries different confidence, and the system can make different decisions depending on the validation state: for example, only including explicitly confirmed translations in exercises, while using auto-verified ones for reading assistance where the stakes are lower.

7.2.3 Problem. How can trusted and untrusted LLM-generated data be told apart after it lands in the database, so each is used according to its confidence?

7.2.4 Forces. Once LLM-generated data is persisted, it becomes indistinguishable from human-verified data unless explicitly marked. Downstream features and users may silently treat unverified AI output as ground truth. Over time, the system accumulates a mix of trusted and untrusted data with no way to tell them apart.

7.2.5 Solution. Maintain an explicit, queryable validation state for all LLM-generated content in the data model. Never let LLM-generated content silently become trusted data. The validation state may be a simple flag, but more often it is a spectrum or state machine reflecting different levels of trust (e.g., unverified → auto-verified → implicitly accepted → explicitly confirmed by user; alternatively, one could track the number of users that have validated a given content).

7.2.6 Consequences.

- **Each consumer can gate on trust.** Exercises use only confirmed pairs, reading assistance can fall back to auto-verified ones, and unverified output never silently becomes ground truth.
- **The data model carries an extra state to maintain.** It must be set on every write and updated on every validation event, and the right granularity (a flag, a spectrum, or an agreement threshold: N independent users confirming before it counts as trusted) is a domain judgment.
- **It adds the second half of the provenance picture.** Where *LLM Output Provenance* records how an artifact was made, this records whether it has been confirmed; the two together answer both questions about any stored artifact.

7.2.7 Notes.

- This pattern complements *LLM Output Provenance*. Together they answer two essential questions about any piece of LLM-generated data: how was it produced? and has anyone confirmed it's correct? The right granularity of validation tracking depends on the domain: some systems may need binary (verified/not), others may need multiple validators with agreement thresholds, and others, like Zeeguu, benefit from a trust spectrum that reflects different forms of implicit and explicit user feedback.
- Implicit validation: One could argue that "if a user practiced a word without complaint, it's implicitly validated"

7.2.8 Known Uses.

- **Argilla** records carry an explicit status lifecycle (pending → draft → submitted, plus validated/discarded), so model suggestions are not trusted until a human validates.
- **Label Studio** tracks a per-task ground-truth flag and review state, distinguishing verified gold data from unreviewed predictions.
- **Snorkel [10]** (Ratner et al., VLDB 2017) attaches probabilistic confidence to programmatically generated labels rather than treating them as ground truth: a confidence spectrum rather than a discrete state machine.
- *Note.* These are data-labeling platforms; a documented, in-product *LLM-output* trust-state lifecycle (as opposed to human-annotation state) remains thinly evidenced: one reason we keep this among the less-settled patterns.

7.3 Hybrid Classical+LLM Pipeline

7.3.1 Context. A task can be served by a fast, deterministic classical tool (a dependency parser, a POS tagger, a rule extractor) that misses edge cases, and by an LLM that handles the edge cases but is too expensive to run on every input.

7.3.2 Example. Multi-word expression (MWE) detection runs Stanza (a classical NLP library) first, as a cheap *high-recall* gate: it catches every possible candidate, tolerating false positives. If Stanza flags no candidate in a sentence, the LLM is never called. When it does flag one, an LLM re-analyzes the whole sentence and makes the *precision* call, rejecting the false positives; its verdict is used even when it overrides Stanza and finds no expression. The LLM therefore runs on only the fraction of sentences that might contain an expression, rather than on every sentence.

7.3.3 Problem. How can both high recall and high precision be reached without paying LLM cost on every input?

7.3.4 Forces. Classical NLP tools (dependency parsers, POS taggers, rule-based extractors) are fast and deterministic but miss edge cases. LLMs handle edge cases well but are expensive to run on every input. Neither alone achieves both high recall and high precision.

7.3.5 Solution. Run the cheap classical tool first, as a high-recall gate: invoke the LLM only when the classical stage fires, and skip it otherwise (the common case, and the main cost saving). When it fires, let the LLM make the precision decision. The two are not alternatives: the classical stage controls *when* the LLM runs; the LLM controls *what counts*.

7.3.6 Consequences.

- **The LLM runs only where it is needed.** It fires on the fraction of inputs the classical gate flags, so the common case costs nothing extra while the LLM still makes the precision call.
- **Two systems must be built and kept in sync.** That is real added complexity, usually justified by the saving from skipping the LLM on the common case.
- **The gate must have high recall.** A candidate the classical stage misses never reaches the LLM, so the recall of the cheap stage caps the precision of the whole.

7.3.7 Note. Close kin to *Escalate to the LLM* and to a model cascade: all three run a cheap step first and call the LLM selectively. The difference is the trigger. Escalate uses the cheap tool's answer and reaches for the LLM only when it is inadequate (a failure, or user dissatisfaction); here the cheap tool gates on detected difficulty (a flagged candidate) and the LLM's verdict replaces it, as in a confidence-based cascade.

7.3.8 Known Uses.

- **RankGPT [11]** (Sun et al., EMNLP 2023) uses BM25 to retrieve ~100 candidates, then an LLM for listwise reranking: classical high-recall generator + LLM precision filter.
- **spaCy-llm** (Explosion) is designed to mix LLM components with classical/rule-based ones in a single pipeline.
- **Retrieve-then-rerank** (e.g. Cohere Rerank) keeps a high-recall first-stage search and adds a neural precision reranker.

7.4 Defensive Output Parsing

7.4.1 Context. An LLM is asked for output in a fixed shape (JSON, a delimited record), but format compliance is only probabilistic, and the parsed result feeds code on the critical path.

7.4.2 Example. Multi-word-expression detection asks the LLM for a JSON array (groups of word positions in a sentence) but refuses to blindly trust that it will get one.

- `_parse_response` first strips any markdown code fence (the model often wraps the JSON in one), tries `json.loads`, and on failure regex-extracts the last JSON array in the text (models tend to add a preamble), parses that, and distinguishes a legitimately empty result (`[]`) from a parse failure.
- If nothing parses, it logs the raw text and returns `[]` rather than raising.
- A separate `_validate_groups` step then checks the parsed shape (token indices in range) before anything downstream uses it.

The same layered try / extract / validate / fall-back appears in the translation validator, the simplification service, and the example generators.

7.4.3 Problem. How can a probabilistic formatting slip be kept from turning into a failed request?

7.4.4 Forces.

- A fixed output format is required (JSON, a delimited record)

- Format compliance is probabilistic
- A naive parse on the critical path can turn a formatting slip into a failed request

7.4.5 Solution. Do not trust the raw output's shape.

Parse in layers: strip known wrappers, attempt a lenient parse, extract the expected structure from the surrounding text if that fails, validate the parsed shape (types, ranges, required fields), and then degrade gracefully (a default, a skip, a retry, or the next provider) instead of raising.

Keep the strictness in code, where it is deterministic and testable, rather than in ever-longer prompt instructions.

7.4.6 Consequences.

- A malformed response degrades a single result instead of failing the request: the layered parse recovers what it can, and a clean fallback (a default, a skip, a retry, the next provider) covers the rest.
- The strictness lives in parsing code that is deterministic and testable, rather than in ever-longer prompt instructions.
- Provider "JSON mode" or function-calling reduces malformed output but does not remove the need to validate the shape before use.

7.4.7 Notes.

- Composes with a one-shot retry and with *Fail-Fast Provider Chain* (a parse failure can trigger the next provider).
- Related to *Deterministic Postprocessing*, which repairs a specific, known formatting defect; this pattern is the broader stance of not trusting the structure at all.

7.4.8 Known Uses.

- **G-Research**'s code-review tool treats LLM output as "unverified input": it normalises responses by "removing wrappers before parsing" (some providers return bare JSON, some wrap it in markdown fences), validates every finding against an authoritative rule index (invalid findings are "rejected outright"), detects truncation via the `length` finish reason and retries with a lower cap, and sends structurally-invalid output back to the model for one repair pass.
- **Honeycomb**'s Query Assistant parses the LLM output, "correct[s] it (if it's correctable)," validates it, and instruments parse vs. validation errors separately before running the query.
- *Tools that ship this.* Validation/repair is widely productized, Instructor (Pydantic + auto-retry), LangChain `OutputFixingParser`, and provider structured-output modes (which still require handling truncation and refusals), but a library that *provides* validation is the mechanism, not evidence of an in-app stance.

7.5 Deterministic Postprocessing

7.5.1 Context. LLM output carries a deterministic formatting defect (a stable trailing string, a known preamble, leaked markdown in a plain-text field), the same defect shows up on every call, and it is already present in rows written to the database.

7.5.2 Example. LLM-simplified article summaries consistently ended with a Unicode ellipsis (…), making every home-card preview read as an unfinished sentence. One option was to add a "do not end with ellipsis" instruction to the simplification prompt; the chosen option was a five-line regex stripping any trailing … or `...+` at serialization time.

Because it runs as each summary is served, not on the stored row, it fixes every case at 100%, including the ~60k rows already in the database, with no backfill and no row ever rewritten.

7.5.3 Problem. Should a deterministic defect be fixed by instructing the model, or in code?

7.5.4 Forces. When LLM output has a deterministic formatting defect (a stable trailing string, a known preamble, leaked markdown in a plain-text field, trailing whitespace), the obvious instinct is to fix it in the prompt. But: - Prompt compliance is probabilistic; the same constraint in code is 100%. - Prompt tokens cost money on every call and can distract the model from the actual semantic task. - Prompt changes do not affect rows already in the database. - Code is testable and reviewable; prompt instructions are not.

7.5.5 Solution. Enforce deterministic constraints in code, at the post-processing or serialization boundary. Reserve prompt instructions for things that genuinely require model judgment.

7.5.6 Consequences.

- A code-side fix is 100% reliable, costs no prompt tokens, is testable and reviewable, and (applied at the serialization boundary) cleans every already-stored row on the way out with no backfill, none of which a prompt instruction achieves.
- It applies only to genuinely deterministic defects. The test is whether the rule needs model judgment: *strip a trailing …* belongs in code, *don't mention the user's name* the model has to enforce. A code-side rule list that keeps growing is a signal the task is poorly scoped, not that it needs more rules.

7.5.7 Known Uses.

- *Adjacent structural cousin.* Structured-output libraries such as **Outlines** constrain generation so the format is valid upfront, and explicitly contrast this with the common practice of "fixing bad outputs after generation using parsing, regex, or fragile code": evidence that code-side repair is widespread, though Outlines *prevents* rather than *repairs*.
- *Largely best-practice / folklore.* We found no source that names this specific rule: repair a *deterministic* defect in code, not the prompt, because code is 100% reliable and also fixes already-stored rows. The literature documents the stronger cousin (constrained decoding / structured outputs) and generic output sanitization, so we present this as its own contribution and cite these as adjacent, not prior.

7.6 Targeted User Feedback

7.6.1 Context. An LLM generates content that users consume directly (a lesson script, example sentences, a translation). Automated validation catches many errors, but some still reach users, and the users reading the output are the best-placed detectors of what slipped through.

7.6.2 Example. Zeeguu lets learners flag LLM-generated content as wrong, in the form that fits each surface:

- **Audio lessons (continuous content).** A lesson script is delivered as audio. When a listener notices a mistake, a **Report** button captures the exact position in the script they were at, so the report points at the specific line rather than the whole lesson.
- **Example sentences (discrete content).** Each generated sentence carries its own report control, so a learner flags the individual sentence that is wrong.

Either way the report says *which* piece of output is wrong, not merely *that* something is.

7.6.3 Problem. How can errors that slip past automated checks be caught and pinpointed, using the users who are already reading the output?

7.6.4 Forces.

- Automated validation (*LLM-Checking-LLM*, classical checks) catches many errors but never all; some reach users.
- The user who hits an error is the cheapest, best-placed detector, but only if reporting is low-friction and in the moment.
- A bare "this is wrong" is hard to act on; a report tied to a specific position or item is directly actionable.
- The right report granularity depends on the content: continuous content (audio, long text) needs a position anchor; discrete content (a sentence, a card) is naturally reported as a unit.

7.6.5 Solution. Give users a low-friction way to flag LLM-generated output, and capture enough context to act on it. Match the report's anchor to the shape of the content: for **continuous** content, anchor to the user's current position (the point in the script); for **discrete** content, attach a per-item report control so the flagged unit is unambiguous.

Route each report into the quality machinery: record it against the artifact (composes with *LLM Content Validation Tracking*), let it trigger regeneration (composes with *Soft Invalidation of LLM Artifacts*), and use *LLM Output Provenance* to identify the prompt and model that produced the flagged output.

7.6.6 Consequences.

- **Users become a last-line error detector.** Mistakes that pass every automated check are still caught, by the people best placed to notice, at almost no cost to the system.
- **A pinpointed report is actionable.** Anchoring to a position or an item turns "something is wrong" into "*this* is wrong," so a report can drive a targeted regeneration instead of a manual hunt.
- **The affordance and the plumbing both have to exist.** A report control has to be designed into each surface, and reports only help if something consumes them (a review queue, an automatic deprecation). A flag nobody acts on is theater.

7.6.7 Notes.

- Collecting user feedback is itself well-known; the facet worth stating is *granularity*: match the anchor to the content (position for continuous, per-item for discrete).
- Explicit reports are the active end of the spectrum whose passive end *LLM Content Validation Tracking* already captures (a user who practiced a word without complaint). Together they run from silent acceptance to active correction.

7.6.8 Known Uses.

- Generic thumbs-up / thumbs-down feedback on LLM output is ubiquitous (ChatGPT, GitHub Copilot, and most assistant UIs collect it). The facet this pattern adds, and that is far less common, is feedback **anchored to a position or item** and wired to **regeneration** of the flagged artifact rather than to offline model training.

8 What Makes These Patterns LLM-Specific?

Some of these patterns echo general distributed systems wisdom (batching, fallback, redundant dispatch). What makes them distinctly relevant to LLM integration is the specific combination of forces:

- **Cost structure**: per-token pricing with high fixed prompt overhead, unlike flat-rate API calls
- **Non-determinism**: same input can yield different outputs, necessitating verification chains
- **Asymmetry between generation and verification**: checking is easier than producing
- **General-purpose capability**: the same component can serve as prototype, primary, or fallback
- **Quality–cost–latency tradeoff space**: uniquely wide compared to traditional APIs

8.1 Relationship to LLM Gateways

Several of these patterns (*Fail-Fast Provider Chain*, *Centralized Model Selection*, and hot-path caching) are being commoditized into LLM-gateway infrastructure (LiteLLM, Portkey, Cloudflare AI Gateway). This does not invalidate them as patterns; it relocates their *implementation* from application code into shared infrastructure. A pattern documents a recurring solution whether it is hand-rolled or shipped by a gateway, and knowing the pattern is precisely what lets one judge whether a given gateway implements it well (e.g. most gateways do *not* provide *Multiplexed Dispatch* with alternative-caching). Connection pooling remained a pattern long after every ORM started shipping one.

The recurring shape across all of these is that the gateway supplies a *mechanism* while the pattern owns the *intent, the domain join, and the decision the mechanism drives*. This is sharpest in *Per-User Consumption Budget*, where the gateway can throttle and cap but only the application knows which resource to bound and what the user should get when the budget is exhausted.

9 Related Work

To our knowledge, no peer-reviewed work presents a catalog of architectural patterns for integrating LLMs as components into existing production systems, grounded in real deployment experience and described using the standard pattern format (context, forces, solution, consequences). Our patterns addressing lifecycle management (Rent, Then Build), cost optimization (Pre-computation, Prompt Amortization, Escalate to the LLM), latency management (Multiplexed Dispatch), quality assurance (LLM-Checking-LLM, Validation Tracking), and data management (Provenance) appear to be novel contributions.

Several practitioner-oriented resources discuss patterns for building LLM-based systems, but they operate at a different level of abstraction and lack the grounding in a specific production system that we aim to provide.

9.1 Practitioner resources

Eugene Yan's "**Patterns for Building LLM-based Systems & Products**" (2023, blog post) identifies seven patterns: Evals, RAG, Fine-tuning, Caching, Guardrails, Defensive UX, and Collecting User Feedback. These address the question "how do I build an LLM product?" They describe the overall stack for LLM-native applications. Our patterns address a different question: "I have an existing system, and I want to add LLM capabilities as a component: how do I manage cost, quality, latency, and lifecycle?" There is some overlap on caching/pre-computation, but our treatment focuses on prompt amortization and user-need anticipation rather than semantic similarity caching. Yan's work is not peer-reviewed.

ThoughtWorks' "**Emerging Patterns in Building GenAI Products**" (Fowler et al., martinfowler.com) and "Engineering Practices for LLM Applications" focus on operational concerns: testing, evaluation, guardrails, and RAG pipelines. These are primarily about LLMOps rather than the architectural decisions for embedding LLMs as components within an existing application.

Andreessen Horowitz's "**Emerging Architectures for LLM Applications**" (2023) provides a reference architecture for the LLM infrastructure stack (embedding pipelines, vector databases, orchestration layers, agents). Again, this targets LLM-native products rather than LLM integration into existing systems.

Books. "**LLM Design Patterns**" (Huang, Packt, 2024) and "LLMs in Enterprise" (Menshawy & Fahmy, Packt, 2025) cover model-level patterns (fine-tuning, quantization, inference optimization, RAG) and enterprise deployment concerns. They do not address application-level integration patterns such as the lifecycle management (Rent, Then Build → specialized tool → Escalate to the LLM), prompt amortization, or LLM output provenance that we identify.

9.2 Academic surveys.

There is a large body of work on **using LLMs *for* software engineering tasks**: code generation, bug repair, testing, requirements engineering (see surveys by Fan et al., 2023; Zhang et al., 2024). However, these focus on LLMs as tools for developers, not on the engineering challenges of integrating LLMs as runtime components within production software.

A recent **systematic literature review on software architecture and LLMs** (Schmid et al., 2025) found only 18 relevant papers and noted that LLM-based software design remains an open research direction. None of the surveyed academic work addresses the specific architectural patterns for managing cost, quality, latency, and lifecycle when LLMs serve as components in existing applications.

LLM self-verification. For the *LLM-Checking-LLM* pattern, there is relevant work on **LLM self-verification**. Gero et al. (2023) demonstrated that self-verification improves clinical information extraction accuracy, explicitly building on the asymmetry between verification and generation. However, Stechly et al. (2024) showed that self-critique fails for formal reasoning tasks, finding significant performance collapse with self-verification but gains with external verification. Our pattern differs from both in that it uses separate, differently-prompted LLM calls for generation and verification of different properties (e.g., text simplification followed by grammatical correction), rather than asking an LLM to verify its own reasoning.

10 Limitations and Future Work

The patterns in this catalogue have been extracted from a single production system, Zeeguu, in the language-learning domain. We have argued throughout that the underlying forces (per-token cost, multi-second latency, non-determinism, a rapidly shifting provider landscape, and general-purpose capability) are not specific to language learning, and that the patterns should therefore generalise to other interactive applications that surface LLM-generated text to end users. Whether this is borne out in practice is an empirical question we cannot fully answer from a single case study. The most direct way to validate or refute the catalogue is to gather complementary examples (and counter-examples) from practitioners working in other domains. A PLoP writers' workshop is one venue for exactly this kind of community refinement; mining open-source repositories for additional instances of these patterns (and patterns we missed) is a natural follow-up.

A second limitation is that the catalogue reports patterns we have found useful but does not yet quantify their impact systematically. Anecdotal cost and latency improvements are reported per pattern; a more rigorous deployment-level

evaluation (measuring, for example, how much of Zeeguu's LLM bill is attributable to which patterns, or how much user-perceived latency *Multiplexed Dispatch* removes) is future work.

A further direction is to treat the catalogue as the seed of a *pattern language* rather than a flat list. The patterns are not independent: they compose (pre-computation feeds *Prompt Amortization*; a report in *Targeted User Feedback* drives *Soft Invalidation of LLM Artifacts*) and they evolve over a system's lifetime (an LLM may begin as a *Rent, Then Build* stand-in, hand off to a specialized tool, and remain as the *Escalate to the LLM* fallback). Documenting these interactions, sequences, and tensions, so a designer can navigate from one pattern to the next, is a contribution beyond the individual patterns.

These choices are not made once. A system's pattern mix is actively revised as it scales and the provider landscape shifts: Zeeguu adopted *Multiplexed Dispatch* only recently, having previously made single LLM calls, and may later drop it or fold it together with *Per-User Consumption Budget* as usage grows. A longitudinal account of how and why a system's chosen patterns change over time would deepen the lifecycle dimension that this single-snapshot catalogue can only sketch.

Finally, we expect the catalogue to be incomplete. Several proto-patterns are listed in *Candidate Patterns*; others will likely surface only through engagement with practitioners working at different scales, in different domains, and with different LLM providers.

11 Conclusion

This paper has presented a catalogue of architectural patterns for integrating LLMs as components into existing user-facing applications: a setting that has received much less attention than the design of LLM-native products, despite arguably affecting a larger share of working software engineers. The patterns address recurring concerns around cost, latency, lifecycle, data management, and quality assurance, and were drawn from real production experience on the Zeeguu platform. They are intended as a starting point for community refinement rather than a closed taxonomy: practitioners working on LLM integration in other domains are warmly invited to validate, refute, extend, or replace individual patterns, and to contribute new ones the catalogue does not yet contain.

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